MATH 15200: Calculus II
Homework 4: Logarithms and Exponentials

Due: Monday, November 9, 2026, at 11:59 PM.

This is the written portion of Homework 4; complete the routine online portion separately in Edfinity. Upload your written solutions to Gradescope, via Canvas. You may collaborate, but you must write and submit your own solutions independently.

1. For each positive integer n, let H_n be the harmonic sum

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

Show that H_n grows logarithmically by proving that

$$\ln(n+1) \le H_n \le 1 + \ln n.$$

Hint: If $k \le t \le k+1$, then $\frac{1}{k+1} \le \frac{1}{t} \le \frac{1}{k}$. Use Problem 2 of Homework 2 to integrate these inequalities over $[k, k+1]$. Then add the results: over $k = 1, \ldots, n$ for the lower bound, and over $k = 1, \ldots, n-1$ for the upper bound.

2. Without using a calculator, determine which is larger, e^π or π^e.

Hint: Since ln is increasing, it is enough to compare $\ln\big(e^\pi\big)$ and $\ln\big(\pi^e\big)$. Rewrite this as a comparison of $g(e)$ and $g(\pi)$, where $g(x) = \frac{\ln x}{x}$ for $x > 0$. Find where g is increasing and where it is decreasing.

3. Let $b > 0$, and let R be the region bounded by the curve $y = e^{-x^2}$, the x-axis, the y-axis, and the line $x = b$. The function e^{-x^2} has no antiderivative given by a formula in familiar functions, so the area of R cannot be computed directly with the Fundamental Theorem of Calculus.

(a) Find the volume $V(b)$ of the solid formed by revolving R about the y-axis.

(b) Find $\lim_{b \to \infty} V(b)$.

Hint: Use the shell method, and then the substitution $u = x^2$.